MILITARY SPECIFICATION

FUZE, ROCKET, M427, MOD 5
LESS BOOSTER AND BOOSTER LEAD

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the parts and assembly drawings of a point detonating delay fuze designated as Fuze, Rocket, M427, MOD 5, Less Booster and Booster Lead (see 6.7).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-D-18665 - Detonator, Mark 59, Mod 0
MIL-D-50866 - Detonator, Stab Delay for Fuze, M427 MOD 5
MIL-P-60412 - Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes; General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For.

FSC: 1340

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MIL-STD-810 - Environmental Test Methods for Aerospace and Ground Equipment.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8886810 - Box, Packing, Ammunition, For Warhead, 2.75 Inch, Rocket, HE, M151, Smoke WP M156, and Practice XM230.
8886811 - Container, Ammo. Fiber, M523, For Warhead, 2.75 Inch Rocket, HE, M151, Smoke WP M156 and Practice XM230
9234553 - Fuze, Rocket, Less Booster and Booster Lead

PUBLICATIONS

ORDNANCE CORPS

ET-9234553 - Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Fuze assemblies and parts.-The fuze assemblies and parts shall comply with all requirements specified on Drawing (dwg.) 9234553, and with all requirements specified in applicable specifications.

3.2.1 Spring embrittlement.-None of the springs used in the fabrication of the fuze shall break when extended or distorted to the point of minimum permanent distortion.

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3.2.2 Spring set.-The setback weight spring shall return within .015 inch of the original free length when compressed and heat treated.

3.2.3 Arming, detent and lock spring functioning.-The rotor assembly in the loaded Safety and Arming Device (S&A) shall rotate from the fully safe position to the fully armed position, the detent shall lock the rotor in the fully armed position, and the lock spring shall block the exit of the detent.

3.3 Arming Time

3.3.1 Safety and arming device.-The rotor assembly in the loaded S&A shall rotate from the fully safe position to the fully armed position in not less than 1.13 sec. and not more than 1.30 sec. when subjected to arming acceleration of 40.00 plus or minus 0.25 g's.

3.3.2 Fuze assembly.-The fuze assembly shall arm in not less than 1.07 sec. and not more than 1.36 sec. when subjected to arming acceleration of 40.00 plus or minus 0.25 g's and the detent shall lock the rotor in the fully armed position as determined by radiographic inspection.

3.4 Non-arming

3.4.1 Safety and arming device.-The rotor assembly in the loaded S&A shall not arm when subjected to an acceleration of 12.75 g's minimum for a minimum of 3 seconds. This test shall be performed just prior to insertion of the S&A into the fuze body.

3.4.2 Fuze assembly.-The fuze assembly shall not arm when subjected to an acceleration of 13.00 g's plus or minus 0.25 g's for a minimum of 3 seconds.

3.5 S&A position.-The safe position of the S&A shall be verified after insertion into the fuze body assembly. After insertion of the S&A fuze handling shall be positively controlled to preclude any chance of inadvertant arming or substitution of the S&A until assembly of the booster housing.

3.6 Jolt and Jumble.-The fuze assembly shall be subjected to and must withstand the Jolt test specified in MIL-STD-331, with the exception that the fuze shall not be disassembled and inspected after the Jolt test. After the completion of the Jolt test, the same fuze shall be subjected to and must withstand the Jumble test specified in MIL-STD-331.

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3.7 Forty-foot drop.-The complete fuze assembly shall withstand the forty-foot drop test specified in MIL-STD-331. To satisfy the requirements of MIL-STD-331, the disassembled fuze must display the following conditions:

a. No loose explosive shall be present in the fuze body assembly.

b. The lock roller pin on the rotor shall not be broken.

c. The rotor shall be in the safe position, secured such that it cannot be turned to the armed position with finger pressure.

d. The g-weight shall be in fully safe position.

3.8 Fuze Assembly Operational Tests.- The unarmed fuzes shall be subjected to and must withstand the following operational tests: Five Foot Drop Test as specified in MIL-STD-331; Non-Environment Test - Non-Arm and Arming Times only; Functional Test - Aircraft Vibration, Method 514, Curve Z, as specified in MIL-STD-810. The vibration conditioning shall be continuously performed until completed. After completion of the Five Foot Drop Test and the Functional Test, including the acceptance criteria for these tests (leak test, non-arming test, and arming time test), the same fuzes shall be tested for high order functioning.

3.9 Seal leakage test.-The leakage in the fuze shall not cause a pressure drop greater than 0.00072 psi in 10 sec.

3.10 Fuze firing.-The fuze assembly shall function the explosive elements high order when a weight is dropped onto the nose of the fuze.

3.11 Workmanship.-All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base material. The loading and assembly operations shall be performed in a thorough workmanlike manner consistent with the best practices of the industry. All required marking shall be neat and sharply defined.

3.12 First article inspection.-This specification makes provisions for first article inspection. Requirement for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.2 Contractor inspection system.-As part of the contractor's approved quality procedure, there shall be a provision for checking the adequacy of inspections made in connection with listed critical or major defects. These procedures shall include the following provisions:

4.1.2.1 Handling procedure.-A Design Agency approved procedure shall be instituted that shall preclude unauthorized handling of the S&A's and fuzes from the S&A position check until the booster housing has been installed. The procedure shall be fail safe to eliminate armed fuzes from being introduced into the lot. Fail safe is defined as follows: Any equipment failure shall operate to cause rejection of the part or otherwise prevent its subsequent use.

4.1.2.2 Identification of rejected material.-All material listed below, rejected as containing defects, shall be isolated to preclude its inclusion with acceptable material:

a. All S&A's failing paragraphs 4.3.3.11, 4.3.3.12, 4.3.3.13, 4.3.3.14 and 4.3.3.15.

b. All fuzes failing paragraphs 4.3.3.16, 4.3.3.17, 4.3.3.18, 4.3.3.19, 4.3.3.20 and 4.3.3.21.

c. All firing pin bodies and striker pin bodies failing paragraph 4.3.3.6.

4.1.2.3 Use of simulated defect standards.-The following listed verified inspections shall be validated and inspection equipment verified hourly using Design Agency approved item facsimilies or standards containing the defects and supplied by the contractor. The contractor shall resupply the item facsimilies when it is determined by the Government that the facsimilies are no longer representative enough of the product to make it unidentifiable to the contractor inspection personnel, or it is not functioning in the necessary manner and cannot be corrected satisfactorily. These standards should be representative of defects actually found or anticipated. They should be inert and readily identifiable to the Government to ensure recovery should they be inadvertently introduced into production. Simulated defect standards used to verify the following verified inspections should be constructed and identified in such a manner that they are not readily recognizable to the contractor inspection personnel as defect standards. Standards should be used so that their number, time, and sequence of introduction into the inspection cycle is known only to those personnel responsible for

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checking the accuracy of the operation. Adequacy of the standards shall be verified at least once each shift.

Verified Inspections

Paragraph 4.3.3.12 - Setback weight return capability.
Paragraph 4.3.3.16 - S&A position check.

NOTE: When there is automatic detection and automatic rejection of defective items to a controlled area or a controlled rejection of defective items, it is permissible for the defect standards to be identifiable and their introduction into production noticeable to contractor inspection personnel.

4.1.2.4 Undetected defect standards.-Provisions shall be established for the corrective action to be taken where the equipment or contractor inspection personnel fail to identify a standard as defective. These provisions shall include a reinspection of all acceptable material subsequent to the last acceptable verification check as a minimum.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of 253 complete fuzes, 25 safety and arming devices loaded, 10 safety and arming devices loaded (Mk 59 detonator only), 25 safety and arming devices unloaded, 25 sets of subassemblies (each set of subassemblies shall consist of every subassembly covered by the drawing package including at least 5 parts from each cavity used to produce plastic parts) and 25 complete sets of component parts (each set shall consist of adequate components to fabricate a complete fuze, except where identical multiple components are used in which case only one per set is required), to a Government approved facility as designated by the Contracting Officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production process,

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procedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials (see 4.1.1.c). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 Inspections to be performed.-Each fuze, subassembly, and component may be examined for every drawing dimension and drawing requirement applicable to the fuze, subassemblies, or components. In addition, any or all of the following examinations and tests will be performed as applicable.

a. 253 complete fuzes as follows:
   (1) 237 fuzes in accordance with 4.3.3.21.
      (a) 45 fuzes in accordance with 4.3.3.21.1
      (b) 96 fuzes in accordance with 4.3.3.21.2, and 4.3.3.21.4.
      (c) 96 fuzes in accordance with 4.3.3.21.3 and 4.3.3.21.4.
   (2) 12 fuzes in accordance with 4.3.3.19.
   (3) 4 fuzes in accordance with 4.3.3.20.

b. 25 safety and arming devices loaded (detonator and primer) in accordance with 4.3.3.11, 4.3.3.12, 4.3.3.13, 4.3.3.14 and 4.3.3.15.

c. 10 safety and arming devices loaded (detonator only) in accordance with 4.3.3.8.

d. 25 sets of subassemblies in accordance with 4.3.3.1, 4.3.3.4, 4.3.3.6,4.3.3.8.

e. 25 sets of component parts in accordance with 4.3.3.2, 4.3.3.3, 4.3.3.5, 4.3.3.6, 4.3.3.7 and the applicable drawings.

4.2.3 Rejection of first article sample.-If any fuze, subassembly, or component fails to comply with the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a fuze, subassembly or component in the sample to comply with any of the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

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4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded fuzes shall contain:

a. Detonators from one interfix lot number from not more than three lots from one manufacturer.

b. Metal parts from one interfix lot number from one manufacturer for each type fuze.

c. Safety and arming devices from not more than two inspection lots for each type fuze.

4.3.1.1 Process specification for plastic parts.-Prior to the start of normal production, the contractor shall obtain approval from the procuring activity for the process specification of plastic parts submitted with the first article sample. This shall specify the time, temperature and pressure ranges which shall be maintained during production of the molded parts. At unscheduled intervals during the operation of the approved process, the Government inspector shall observe the process to ascertain that the approved limits are being maintained. No deviation from these limits without approval from the procuring activity shall be allowed. In the event of an approved deviation, the contractor shall submit a first article sample.

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4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Body, striker pin (see dwg. 8883734 covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Wall thickness, min	Gage	01001
Minor:	AQL 1.5 percent		
201.	Depth of firing pin body cavity, max	Gage	01002
202.	Diameter of firing pin body cavity, max	Gage	01003
203.	Concentricity of nose with diameter of firing pin body cavity	Gage	01004
204.	Evidence of poor workmanship (see 3.11)	Visual	01005

4.3.2.2 Body, firing pin (see dwg. 8883741 covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 1.5 percent		
101.	Thickness at nose wall	Gage	02001
102.	Concentricity of minor diameter of firing pin nut thread with hammer cavity diameter	Gage	02002
103.	Concentricity of hammer cavity diameter with striker pin body assembly mating diameter	Gage	02003
104.	Concentricity of booster cavity diameter with hammer cavity diameter	Gage	02004
105.	Concentricity of booster cavity pitch diameter with warhead mounting pitch diameter	Gage	02005
106.	Pitch diameter of warhead mounting thread	Gage	02006
107.	Length from aft face to warhead mounting shoulder, max	Gage	02007
108.	Depth of booster cavity	Gage	02008
109.	Diameter of S&A cavity	Gage	02009
110.	Concentricity of S&A cavity with booster cavity	Gage	02010
111.	Minor diameter of booster cavity thread, max	Gage	02011
112.	Diameter of "o" ring seat	Gage	02012
113.	Concentricity of "o" ring seat to booster cavity pitch diameter	Gage	02013
114.	Length of "o" ring seat from aft face	Gage	02014
Minor:	AQL 2.50 percent		
201.	Major diameter of warhead mounting thread, min	Gage	02015
202.	Width of wrench slots, min	Gage	02016
203.	Width across wrench slots, max	Gage	02017
204.	Width of machine service groove, min	Gage	02018
205.	Diameter of machine service groove	Gage	02019
206.	Length from warhead mounting shoulder to machine service groove	Gage	02020
207.	Evidence of poor workmanship (see 3.11)	Visual	02021

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4.3.2.3 Pin, striker (see dwg. 8883735, covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Length of conical section	Gage	03001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	03002

4.3.2.4 Anti-setback washer (see dwg. 8883737, covering a detail of 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.5 percent		
201.	Distance between tabs	Gage	04001
202.	Width of tabs, min	Gage	04002
203.	Thickness	Gage	04003
204.	Evidence of poor workmanship (see 3.11)	Visual	04004

4.3.2.5 Sleeve, firing pin (see dwg. 8883738, covering a detail of 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined			
Minor:	AQL 1.0 percent		
201.	Concentricity of outside diameter with inside diameter	Gage	05001
202.	Evidence of poor workmanship (see 3.11)	Visual	05002

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4.3.2.6 Pin, firing (see dwg. 8883739, covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Overall length, min.	Gage	06001
102.	Small outside diameter	Gage	06002
103.	Radius of firing pin point, max.	Gage	06003
104.	Concentricity of diameter at basic point with large outside diameter	Gage	06004
105.	Length of firing pin head, min.	Gage	06005
106.	Diameter of point at basic length	Gage	06006
Minor:	AQL 2.5 percent		
201.	Inside diameter, max	Gage	06007
202.	Distance from tip of firing pin to bottom of cavity	Gage	06008
203.	Large outside diameter, min.	Gage	06009
204.	Damaged or incomplete point	Visual	06010
205.	Evidence of poor workmanship (see 3.11)	Visual	06011

4.3.2.7 Nut, firing pin (see dwg. 8883740, covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Inside diameter, min.	Gage	07001
102.	Concentricity of inside diameter with major diameter of thread	Gage	07002
Minor:	AQL 1.0 percent		
201.	Outside diameter (at top of firing pin nut)	Gage	07003
202.	Evidence of poor workmanship (see 3.11)	Visual	07004

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4.3.2.8 Hammer, firing pin (see dwg. 8883742, covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Total length, min.	Gage	08001
102.	Large outside diameter, min.	Gage	08002
103.	Small outside diameter, min.	Gage	08003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	08004

4.3.2.9 Housing, booster (see dwg. 8883682 covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.5 percent		
101.	Total length	Gage	09001
102.	Location of counterbore from bottom of booster cavity	Gage	09002
103.	Diameter of booster cavity	Gage	09003
104.	Diameter of lead cavity	Gage	09004
105.	Depth of lead cavity	Gage	09005
106.	Height of lead	Gage	09006
107.	Thickness at end of lead cavity	Gage	09007
108.	Width of O-ring groove	Gage	09008
109.	Diameter of O-ring groove	Gage	09009
110.	Concentricity of thread pitch diameter with booster cavity	Gage	09010
111.	Concentricity of lead cavity with booster cavity	Gage	09011
112.	Major diameter of threads	Gage	09012
113.	Concentricity of O-ring groove with pitch diameter	Gage	09013
114.	Finish of O-ring groove	Visual	09014

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Minor:	AQL 2.5 percent		
201.	Width across wrench slots	Gage	09015
202.	Symmetry of wrench slots	Gage	09016
203.	Inside diameter at crimp section	Gage	09017
204.	Wall thickness at crimp section	Gage	09018
205.	Protective finish missing or incomplete	Visual	09019
206.	Evidence of poor workmanship (see 3.11)	Visual	09020

4.3.2.10 Pad, booster (see dwg. 9234461 covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Thickness	Gage	10001
Minor:	AQL 1.0 percent		
201.	Diameter, max	Gage	10002
202.	Evidence of poor workmanship (see 3.11)	Visual	10003

4.3.2.11 Disc, closing, booster (see dwg. 8883681 covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.5 percent		
201.	Diameter	Gage	11001
202.	Thickness	Gage	11002
203.	Evidence of poor workmanship (see 3.11)	Visual	11003

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4.3.2.12 Striker pin body assembly (see dwg. 8883733 covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Concentricity of striker pin with diameter of firing pin body cavity	Gage	12001
102.	Striker pin loose	Visual/Manual	12002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	12003

4.3.2.13 Firing pin body assembly (see dwg. 8883736 covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Depth of safety arming device cavity, min (after staking)	Gage	13001
102.	Anti-setback washer improperly centered in firing pin cavity	Visual	13002
103.	Firing pin improperly aligned	Visual	13004
Minor:	AQL 1.0 percent		
201.	Firing pin nut improperly staked	Visual	13004
202.	Evidence of poor workmanship (see 3.11)	Visual	13005

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4.3.2.14 Device, Safety and Arming (see dwg. 9234451 covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical: 100 percent			
1.	Distance between annular gear and No. 2 gear max	Gage	14001
Major:	AQL 0.40 percent		
101.	Alignment of detonator cavity with the outside diameter across the rotor housing posts	Gage	14002
Minor: None defined.			

4.3.2.15 Device, Safety and Arming (see dwg. 9234552 covering a detail of dwg. 9234553).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.5 percent		
101.	Height from forward face of base to top of rotor, max	Gage	15001
102.	Distance from rotor flat to edge of base, min	Gage	15002
103.	Delay detonator inverted	Visual	15003
104.	Delay detonator not staked securely in place	Visual	15004
105.	Delay detonator perforated or deformed by stake	Visual	15005
106.	Detonator inverted (the yellow disc should be observed through aft end)	Visual	15006
107.	Detonator perforated or deformed by stake	Visual	15007
108.	Foreign matter present	Visual	15008
109.	Lubricant missing	Visual	15009

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110.	Diameter across posts of rotor housing, min.....Gage		15010
111.	Protective finish missing or incomplete (all visible plated parts excluding wear and staked areas).....Visual		15011
112.	Markings missing or incomplete.....Visual		15012
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11).....Visual		15013

4.3.2.16 Fuze, Rocket, M427, MOD 5 less booster and booster lead (see dwg. 9234553 as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.0 percent		
101.	Gap between striker pin body and firing pin body, max.....Gage		16001
102.	Depth of crimp, min.....Gage		16002
103.	Booster housing improperly torqued.....Gage		16003
104.	Length from aft face to end of booster housing max.....Gage		16004
105.	Protective finish incomplete or inadequate on outside surfaces except in stamping areas & crimping surface.....Visual		16005
106.	Stamping missing or incomplete.....Visual		16006
Minor	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11).....Visual		16007

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4.3.3 Testing

4.3.3.1 Torque test of firing pin nut (see dwg. 8883736) - Major defect - Code No. 13006.-The sampling plan for this test shall be in accordance with MIL-STD-105 using inspection level II with an AQL of 0.40 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.2 Spring embrittlement (see 3.2.1) - Major defects; detent spring - Code No. 17001 - set-back weight spring - Code No. 18001; lock spring - Code No. 19001.-A sample of 10 springs shall be selected from each heat treatment batch. The sample shall be tested as specified in 4.4.2 using equipment in accordance with 4.3.4. The heat treatment batch shall be rejected if one or more springs fail to comply with the specified requirement. Springs tested as specified shall be discarded as scrap.

4.3.3.3 Setback weight spring set (see 3.2.2) - Major defect - Code No. 18002.-A sample of ten (10) setback weight springs shall be selected from each heat treatment lot and shall be tested as specified in 4.4.3 using equipment in accordance with 4.3.4. If one or more springs fail to comply with the specified requirement they shall be classed defective and the heat treatment lot shall be rejected. Springs tested as specified shall be discarded as scrap.

4.3.3.4 Mechanical Tests

4.3.3.4.1 Critical defects.-These tests shall be performed 100 percent. Assemblies failing to comply with the specified requirements shall be classed defective and removed from the lot. The tests shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

a. Load test of No. 1 pinion (see dwg. 8883703) Code No. 20001

b. Load test of No. 2 pinion (see dwg. 8883700) Code No. 21001

c. Load test of escapement pinion (see dwg. 8883710) Code No. 22001

d. Load test of lever shaft (see dwg. 8883706) Code No. 23001

e. Load test of lever pallet (see dwg. 8883706) Code No. 23002

f. Load test of annular gear (see dwg. 9234550). Code No. 24001

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4.3.3.4.2 Critical defects.- The sampling plan for these tests shall be in accordance with MIL-STD-105 using inspection level II and an AQL of 0.040 percent for each defect. Continuous sampling may be used in accordance with MIL-STD-1235 using inspection level II, CSP-I and an AQL of 0.015 percent for each defect. The tests shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

a. Load test of lock roller pin (see dwg. 9234550) Code No. 24002

b. Torque test of lever shaft (see dwg. 8883706) Code No. 23003

c. Load test of bridge rivet (see dwg. 8883693) Code No. 25001

4.3.3.4.3 Major defects.- The sampling plan for these tests shall be in accordance with MIL-STD-105, using inspection level II and an AQL of 0.40 percent for each defect. The tests shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

a. Load test of spring stud (see dwg. 8883719) Code No. 26001

b. Load test of pillars (see dwg. 8883692) Code No. 27001

c. Torque test of rotor stop stud (eccentric pin) (see dwg. 9234552) Code No. 14003.

4.3.3.5 Rotor Housing Arm Width - (see dwg. 8883719) - Major defect - Code No. 28001.-This test shall be performed 100 percent. Any rotor housing assembly found with a rotor housing arm width (arm on the side nearest the studs) in excess of the drawing requirement at any point over the entire length shall be classed defective and removed from the lot. The test shall be performed with equipment in accordance with 4.3.4.

4.3.3.6 Fuze Body component dimensional examination - Firing pin body length from nose to shoulder (see dwg. 8883741) - Major defect - Code No. 02022. Striker pin body assembly length from point of striker pin to end of striker pin body (see dwg. 8883733) - Major defect - Code No. 12004.-These tests shall be performed 100 percent. Any component failing to meet the specified requirement shall be classed defective and removed from the lot in accordance with 4.1.2.2. The equipment to perform these tests shall be in accordance with 4.3.4.

4.3.3.7 Dimensional examination of inside radius at bottom of nose section (see dwg. 8883734) - Major defect - Code No. 01006. A sample of 20 components shall be selected from each lot of bodies, if one or more bodies fail to comply with the specified requirement the lot shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4.

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4.3.3.8 Testing of Anodic Coating.-The testing of anodic coating shall be in accordance with MIL-A-8625 except that panels shall not be used and paragraph 4.3, "Lot" and 3.10, "Detail Requirements" shall be deleted and the following substituted:

Corrosion resistance of anodic coating - Major defect - Code No. 29001.-The samples of fuze body assemblies (see dwg. 9215611) for corrosion resistance shall be selected in accordance with MIL-STD-105 using inspection level S-2. Lot size for each fuze body part shall be of not more than 40 hours of anodic production. If one or more of the samples fail the requirements the lot shall be rejected. The samples shall be tested as specified in accordance with 4.4.6 using equipment in accordance with 4.3.4. Components tested as specified shall be discarded as scrap.

4.3.3.9 Detonator push test (see dwg. 9234552) - Major defect - Code No. 15014.-The sampling plan for this test shall be in accordance with MIL-STD-105 using inspection level II and an AQL of 0.40 percent. This test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.10 Check test for deterioration of detonators.-If the total time between original acceptance of any detonator lot and the assembly of that lot into the fuzes exceeds two years, at any time since previous tests, the detonator lot shall be subjected to, and must satisfactorily pass, the functioning and output tests specified in MIL-D-18665 and MIL-D-50866 immediately before the detonator lot is assembled into the fuzes. This test shall be performed by the contractor on detonators selected by the Government Inspector at the facility assembling the detonators into the rotors. The test shall be conducted at Government expense without cost to the contractor who loaded the detonators or the contractor who assembled the detonators into the fuzes. The test shall not consitute a basis for rejection against either contractor except where deterioriation has occurred as a direct result of carelessness in handling, storage, etc., permitted while the detonator lot was under the jurisdiction of either contractor.

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4.3.3.11 Arming - Code No. 15015; Detent functioning - Code No. 15016; Lock spring functioning - Code No. 15017 - (see 3.2.3) Major defects.-These tests shall be performed 100 percent on the loaded S&A. S&A's failing to comply with the specified requirements shall be classed defective and removed from the lot. The tests shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4.

4.3.3.12 Setback weight return capability (see dwg. 9234552 Critical defect - Code No. 15018.-This test shall be performed 100 percent on the loaded S&A. S&A's failing to comply with the specified requirements shall be classed defective and removed from the lot. Defect standards for verification of the inspection process shall be in accordance with 4.1.2.3. Defect standards for use with automatic equipment shall be as follows:

a. Units with setback weights which bind when depressed and remain in the depressed condition.
b. Units with setback weights which start to return but do not completely return to the original locking position.
c. Units with setback weights which return from the depressed condition at a rate slower than the normal rate causing rejection.

In the event of automatic equipment breakdown and substitution of manual equipment, defect standards in accordance with "a" and "b" above are required. Failure to detect defect standards shall put in effect procedures in accordance with 4.1.2.4. The test shall be performed with equipment in accordance with 4.3.4.

4.3.3.13 Rotor reset capability (see dwg. 9234552) - Critical defect - Code No. 15019.-This test shall be performed 100 percent on the loaded S&A. S&A's failing to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4.

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4.3.3.14 Arming time (see 3.3.1) - Major defect - Code No. 15020.-This test shall be performed 100 percent on the loaded S&A. S&A's failing to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.10 using equipment in accordance with 4.3.4.

4.3.3.15 Non-arming (see 3.4.1) - Critical defect Code No. 15021.-This test shall be performed 100 percent just prior to insertion of S&A into the fuze body. S&A's failing to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.11 using equipment in accordance with 4.3.4.

4.3.3.16 S&A position check (see 3.5) - Critical defect - Code No. 15022.-The S&A position, after insertion into the fuze body but prior to assembly of the booster housing, shall be inspected 100 percent. Any armed fuze shall be considered defective and removed from the lot and the assembly operation shall be halted until operation is corrected to Government satisfaction. The inspection shall be performed to the requirements of 3.5 using procedures in accordance with 4.1.2 and equipment in accordance with with 4.3.4.

Defect standards with semi-armed and fully armed conditions shall be introduced at various times to verify test performance (see 4.1.2.3). The Design Agency approved handling procedure to preclude untested S&A's from being introduced into the fuze prior to the booster housing being installed shall also be assessed when the standards are introduced. If an improper procedure is being used as determined by the Government, all fuzes accepted subsequent to the last acceptable handling procedure assessment shall be screened as determined by the Government.

4.3.3.17 Fuze marking to verify S&A safe position - Major defect - Code No. 16008.-This inspection shall be performed 100 percent just prior to packing. Each fuze shall be visually inspected to determine that the fuze is color coded on the nose end of the fuze, verifying compliance with the S&A position check. Any fuze with the mark missing shall be considered defective and removed from the lot.

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4.3.3.18 Seal leakage test (see 3.9) - Major defect - Code No. 16009.-The seal leakage test shall be performed on a sample of fuze assemblies selected from each lot in accordance with MIL-STD-105 using inspection level II and an AQL of 0.40 percent. This test shall be performed as specified in 4.4.12 using equipment in accordance with 4.3.4.

4.3.3.19 Jolt - Code No. 16010, and Jumble - Code No. 16011 (see 3.6) - Critical defects.-A sample of twelve (12) fuze assemblies shall be selected from each lot and be subjected to the test as specified in 4.4.13 using equipment in accordance with 4.3.4. If one or more fuze assemblies fail to comply with the specified requirements, the lot shall be rejected.

4.3.3.20 Forty-foot drop test (see 3.7) - Critical defect - Code No. 16012.-A sample of four (4) fuze assemblies shall be selected as part of the First Article Sample. The fuze assemblies shall be subjected to the test as specified in 4.4.14 using equipment in accordance with 4.3.4.

4.3.3.21 Fuze Assembly Operational Tests (see 3.8).-A random sample of forty-five (45) fuze assemblies shall be selected for the Five Foot Drop Test as required in 4.3.3.21.1. A random sample of 315 fuze assemblies, in accordance with MIL-STD-105, sample size M, shall be selected from each lot as the total combined sample for the Non-Environment Test and the Functional Test. Operability of the fuze assemblies in the Non-Environment Test and the Functional Test shall be indicated by subjection to the tests and acceptance or rejection criteria in 4.3.3.21.4. Testing for operability under 4.3.3.21.1 and 4.3.3.21.4 shall be as specified in 4.4.10 and 4.4.11, the firing test shall be performed as specified in 4.4.15 using equipment in accordance with 4.3.4.

4.3.3.21.1 The Five Foot Drop Test, as specified in MIL-STD-331, shall be performed with the following conditions:

a. A sample of 45 fuzes selected in accordance with 4.3.3.21 shall be subjected to the following test.

b. The fuze assembly will be assembled to a MK1 warhead (The Government shall supply the warhead for the contractors use), Filler E loaded, and with impact orientation of the warhead fuze assemblies for the drop test as follows:

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(1) Major axis horizontal

(2) Major axis 45 degrees from vertical base down

(3) Base down

Fifteen fuzes each must be dropped in the above (1,2,3) orientations. No fuze shall be subjected to more than one drop. The orientation of the safety and arming device (specifically the g-weight) of the fuze, shall be random.

c. The Five Foot Drop Test, as specified in MIL-STD-331, shall be performed until three (3) consecutive lots (a minimum lot size of one week's production) have complied with the acceptance criteria specified. After satisfying the above requirement one (1) lot (a minimum lot size of one week's production) per month shall be selected by the Government and subjected to and pass the Five Foot Drop Test. Should the one lot (per month) fail the acceptance criteria, the contractor shall be required to return to the three (3) consecutive lot procedures above for process qualification.

d. The fuzes shall be subjected to the tests and acceptance or rejection criteria as specified in Table I.

Table I

Sample Size 45

Test	Accept	Reject	Defect	Code No.
Non-Arm	0	1	Critical	16013
Arming Time				
Under Min	0	1	Critical	16014
Over Max	2	3	Major	16015
Fail to Arm	1	2	Major	16016
Firing	1	2	Major	16017

If the sample fails the above acceptance criteria the lot shall be rejected.

4.3.3.21.2 Non-Environment Test.- A sample of 219 fuze assemblies shall be subjected to the tests and acceptance or rejection criteria as specified in 4.3.3.21.4. This sample may be selected prior to application of the sealing compound to the booster housing threads and may be assembled without sealing compound.

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4.3.3.21.3 Functional Test.- A sample of 96 fuzes shall be subjected to Aircraft Vibration, Method 514, Curve Z as specified in MIL-STD-810. Vibration conditioning at resonance is not required. After vibration the fuzes shall be subjected to the Seal Leakage Test, 4.3.3.18, and the lot shall be rejected if four (4) or more defects occur. Following seal leakage testing the fuzes shall be subjected to the tests and acceptance or rejection criteria specified in 4.3.3.21.4.

4.3.3.21.4 Acceptance Criteria.- The sample fuzes from the Non-Environment Test and the Functional Test shall be combined into one sample and subjected to the Non-Arm and Arming Time Tests. If one or more fuzes fail to comply with the Non-Arm requirement (see 3.4) - Critical defect - Code No. 16013 - or the Minimum Arming Time requirements (see 3.3) - Critical defect - Code No. 16014 - the lot shall be rejected. The Arming Time defects classified as Major defects shall be accepted in accordance with Table II.

TABLE II

Sample Size 315

DEFECT	CODE NO.	POINTS	ACCEPT	REJECT
Arming Time Over Max.	16015	1	5 POINTS	6 POINTS
Failure To Arm	16016	3		

Acceptable fuzes from the 96 sample Functional Test shall be subjected to firing (see 3.10) - Major defect - Code No. 16017 - after the Arming Time Test. If two (2) or more samples fail to fire the lot shall be rejected. If all tests are acceptable the sample fuzes from the Non-Environment Test may be returned to the lot after disassembly, resetting the S&A into the safe position and reassembly with sealing compound.

4.3.3.21.5 Definitions

a. Times under 1.07 seconds are classed as defect Code No. 16014.

b. Times 1.37 seconds through 1.42 seconds are classed as defect Code No. 16015.

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c. Times over 1.42 and Failures to Arm are classed as defect Code No. 16016.

d. Identification of defects under Code Nos. 16013, 16014, 16015, 16016 will depend on both centrifuge testing and radiographic examination.

e. Any fuze classified defective as per defect 16013, 16014, 16015, 16016 shall be removed and not subjected to firing (see 6.6).

f. All fuzes not classified defective as per defect 16013, 16014, 16015, 16016 shall be radiographically inspected prior to firing.

g. NOTE: If the test equipment fails to produce an arming readout (i.e, recording paper skewes, audio pick-up or amplifier inoperative) during the arming time test as verified by the Government, the safety and arming device shall be removed from the fuze body, and if found to be armed or partially armed the safety and arming device shall be reset into the unarmed position, reassembled into the same fuze body and retimed once as a fuze. This test shall be performed to the timing criteria of this paragraph. The only handling action performed on the safety and arming device shall be the reset of the device just prior to the arming time test. The timing record produced by the fuze on the approved fuze timing equipment shall be considered the timing record of the fuze under the applicable defect classification (16014, 16015, and 16016). Safety and arming devices removed from the fuze body and found to be unarmed shall be classified defective per defect 16016.

h. Fuze arming times, if recorded to the thousandth of a second, shall be "rounded off" to the nearest hundredth of a second. Example: a. 1.065 shall be 1.06
1.066 shall be 1.07
b. 1.365 shall be 1.36
1.366 shall be 1.37

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4.3.4 Inspection Equipment.-Equipment Tabulation Number ET 9234553 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Torque test of firing pin nut (prior to staking).- Torque shall be applied between the firing pin nut and the firing pin body. The resistive force shall act on the firing pin body.

4.4.2 Spring embrittlement.-The spring shall be stretched along its longitudinal axis, in the case of a coil spring, or bent in the case of a flat spring, until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classed defective.

4.4.3 Setback weight spring set.- The springs shall be meaured at their original free length, assembled in a restraining fixture in accordance with 4.3.4 and compressed to a normal height of .750 inches, placed in an oven and subjected to a temperature of not more than 180 degrees Fahrenheit (F.) or less than 175 degrees F. for 24 hours. The springs shall be removed from the oven and be cooled to ambient temperature. After cooling the springs for 24 hours, min. the springs shall be measured and the measurement shall be compared to the original free length to ascertain compliance with the specified requirement.

4.4.4 S&A Mechanical Tests -

4.4.4.1 Load test of No. 1 gear and pinion assembly. - Apply an axial load perpendicular to the No. 1 gear and in such a direction as to test the assembly of the No. 1 gear and pinion. The resistive force shall act against the No. 1 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement.

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4.4.4.2 Load test of No. 2 gear and pinion assembly.- Apply an axial load perpendicular to the No. 2 gear and in such a direction as to test the assembly of the No. 2 gear and pinion. The resistive force shall act against the No. 2 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement.

4.4.4.3 Load test of escapement gear and pinion assembly.- Apply an axial load perpendicular to the escapement gear and in such a direction as to test the assembly of the escapement gear and pinion. The resistive force shall act against the escapement gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement.

4.4.4.4 Load test of lever shaft.- Apply an axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and lever shaft. The resistive force shall act against the lever. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement.

4.4.4.5 Load test of lever pallet.- Apply an axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and the lever pallet. The resistive force shall act against the lever. Upon removal of the required force, the height of the lever pallet above the lever shall be inspected for compliance with the drawing requirement.

4.4.4.6 Load test of annular gear.- Apply a load perpendicular to the rotor and in such a direction as to test the assembly of the annular gear studs to the annular gear and the rotor. The resistive force shall act against the rotor. Upon removal of the required force, the distance from the face of the rotor to the outside face of the annular gear shall be inspected for compliance with the drawing requirement.

4.4.4.7 Load test of lock roller pin.- Apply an axial load perpendicular to the rotor and in such a direction as to test the strength of the assembly of the lock roller pin to the rotor. The resistive force shall act against the rotor.

4.4.4.8 Torque Test of Lever Shaft.- A Torque shall be applied between the lever and the lever shaft.

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4.4.4.9 Load test of bridge rivet.- Apply the required axial load perpendicular to the inner plate and into the bridge rivet from the side opposite the bridge. The resistive force shall act against the inner plate.

4.4.4.10 Load test of spring stud.- Apply an axial load perpendicular to the rotor housing and in such a direction as to test the assembly of the spring stud and the rotor housing. The resistive force shall act against the rotor housing.

4.4.4.11 Load test of pillars.- Apply the required axial load perpendicular to the inner plate and in such a direction as to test the assembly of the pillars to the inner plate and outer plate. The resistive force shall act against the inner plate.

4.4.4.12 Torque test of rotor stop stud (eccentric pin).- The required torque shall be applied between the rotor stop stud and the gear assembly. The resistive force shall act on the gear assembly.

4.4.5 Dimensional examination of inside radius at bottom of nose section.- The component shall be sectioned through the longitudinal axis and the inside radius at bottom of nose section shall be measured. A cast of the radius may be made and gaged on an optical comparator in lieu of sectioning.

4.4.6 Corrosion resistance of anodic coating.- The testing of the sample shall be in accordance with the procedures specified in MIL-A-8625 except that the salt spray test shall be conducted for 168 hours. The significant area for examination shall be the outside surface of the speciments. The acceptance criteria shall be:

a. The specimens shall show no more than 5 spots or pits, none larger than 1/32 inch in diameter, except in those areas within 1/16 inch from the edges, corner, identification markings, and holding during processing.

b. Powdery areas or loose films in excess of 10 percent of the examined surface shall be cause for rejection.

4.4.7 Load test of detonator. - Apply the required axial load perpendicular to the rotor and in such a direction as to test the strength of the staking of the detonator in the rotor.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

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4.4.8 Arming, detent functioning and lock spring functioning.- The S&A in the safe position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the safety and arming device to 20.75 g's maximum for a minimum of 3 plus 1, minus 0 seconds in a direction which simulates rocket launch. At the completion of acceleration, the equipment shall be halted and visual observation shall be made for the locking position of the detent and blocking position of the lock spring. Finger tip pressure shall be applied to the rotor to rotate in the safe direction to ascertain positive arming. At the completion of the test the safety and arming device shall be returned to the safe position.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.4.9 Rotor reset capability.-The loaded S&A shall be assembled to the test fixture in accordance with 4.3.4. The rotor shall be rotated to a position at least 26 degrees but not more than 35 degrees from the fully safe position and released. Observation shall be made for rotor return to the fully safe position.

4.4.10 Arming time

4.4.10.1 Loaded safety and arming device.-The loaded S&A with the g-weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the safety and arming device in a clockwise direction to 40.00 g's plus or minus 0.25 g's and the arming time shall be recorded to the nearest hundredth of a second using a calibrated timer.

4.4.10.2 Fuze assembly.-The fuze assembly with the g-weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the fuze in a clockwise direction to 40.00 g's plus or minus 0.25 g's and the arming time shall be recorded to the nearest hundredth of a second using a swing fixture and acoustic pickup feeding the arming cycle noise signal to a design agency approved recorder.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

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4.4.11 Non-arming

4.4.11.1 Safety and arming device.-The loaded S&A with g-weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the safety and arming device in a clockwise direction to 12.75 g's minimum for a minimum of 3 seconds. Observation shall be made for rotor in the safe position.

4.4.11.2 Fuze assembly.-The fuze assembly with the g-weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the assembly in a clockwise direction to 13.00 g's plus or minus 0.25 g's for a minimum of 3 seconds. X-Ray examination shall be made for the rotor in the safe position.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.4.12 Seal leakage test.- The fuze assembly shall be secured to the applicable test fixture and an air pressure of 5 plus or minus 0.25 psi applied to the base of the fuze. The integrity of the seal shall be verified by measuring the pressure drop per unit time. A leak in the fuze seal causing a pressure drop in excess of 0.00072 psi in 10 sec. shall indicate excessive leakage and reject the fuze. The test equipment calibration and equipment sensitivity shall be in accordance with 4.3.4.

4.4.13 Jolt and Jumble.-The fuze assembly shall be subjected to the Jolt test specified in Standard MIL-STD-331 with the exception that the fuze shall not be diasssembled and inspected after the Jolt test. After completion of the Jolt test the same fuze shall be subjected to and withstand the Jumble test specified in MIL-STD-331.

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4.4.14 Forty Foot Drop.-The fuze assembly with warhead assembly shall be subjected to the Forty Foot Drop test as specified in MIL-STD-331 with the exception that the complete round shall be dropped base down only in a complete pack. The pack is as specified on dwgs. 8886810 and 8886811.

4.4.15 Firing of fuze assembly.-The armed fuze shall be installed in the test fixture in accordance with 4.3.4 and a mass impact of 50 to 51 lbs. shall be dropped from a height of 5.0 to 5.5 ft. onto the nose of the fuze. Suitable stops shall be utilized to prevent crushing of the safety and arming device. The weight shall be dropped only once.

5. PREPARATION FOR DELIVERY

5.1 Packaging and marking - Level C.-The fuze, less booster assembly shall be packed and marked in accordance with MIL-P-60412. Marking shall be on labels.

5.2 Packing and marking - Level C.-The packaged fuzes shall be packed and marked in accordance with MIL-P-60412. Marking shall be on labels. The ICC nomenclature shall be "Percussion Fuzes".

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5. Equipment Lists for the M423/M427 fuze, which are applicable to the M427 Mod 5 fuze, may be employed for the inspection of similar defects.

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6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the neces-

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sary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND9. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of First Article Samples.-Instructions as to the location for the evaluation of the first article sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquires should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND9, Dover, New Jersey 07801.

6.5 Government furnished parts.

6.5.1 Packing and packaging.-The procuring agency will provide the packing and packaging for the Forty Foot Drop Test.

6.5.2 M151 Warheads.- The procuring agency will provide inert loaded M151 warheads for the Forty Foot Drop Test.

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6.5.3 MK1 Warheads.- The procuring agency will provide inert loaded MK1 warheads for the 5 Foot Drop Test.

6.6 Defect reporting.-From each lot produced, the contractor shall submit two copies of the Metal Parts Lot Acceptance Test Data, to Picatinny Arsenal, ATTN: SMUPA-ND9 and SMUPA-DJ3; within three (3) days after completion of acceptance testing of the lot. If fuzes and/or tests in addition to acceptance requirements are utilized to evaluate the lot, the results of these tests will also be submitted three (3) days after the additional tests are completed. The defective fuzes shall be saved pending review of the results.

6.7 Intended use.-It is intended that the item of this specification will be used in the 2.75 Inch Rocket.

Custodian:
Army-MU

Preparing activity:
Army-MU

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